

3(a)	<i>any four from:</i> 1 disaccharide to monosaccharides ; 2 R is water / water required ; 3 hydrolysis reaction ; 4 glycosidic bond broken ; 5 S is α-glucose ; 6 detail of enzyme action ; e.g. induced fit or lock and key hypothesis described 7 AVP ; e.g. (β-)1,4(-glycosidic) bond broken	4
3(b)(i)	<i>intracellular advantage:</i> <i>idea of control / maintaining balance / efficient metabolism ;</i> e.g. if, (enough) glucose / galactose / monosaccharides, present then no need for, uptake / breakdown, of lactose avoids osmotic problems as no build-up of monosaccharides high quantities of product means no requirement for use in respiration by cell <i>disadvantage:</i> loss of product / reduced productivity / product required continuously / slows rate of reaction / ref. to enzyme needing to remain active ;	2
3(b)(ii)	<i>any one from:</i> products and enzyme kept separated / AW ; product removed immediately ;	1
3(b)(iii)	<i>any one from:</i> inert / unreactive / cannot be digested by lactase / AW ; non-toxic ; insoluble ; long shelf-life ; AVP ; e.g. can create small mesh size suggestion of enzyme attachment to fibres	1

Question	Answer	Marks
3(c)	any three from: 1 ref. to controlled variables ; e.g. constant, pH / temperature 2 take samples at timed intervals ; A regular intervals 3 determine, substrate concentration / product concentration ; 4 plot graph of, dependent variable (y-axis) against time (x-axis) ; 5 ref. to rate of disappearance of substrate or ref. to rate of appearance of product ; 6 determine initial rate ;	3

Question	Answer	Marks
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